AN ELEMENT CUTTING THE SUPPORT DOWN BY ONE ACTS AS A NONZERODIVISOR ON A COHEN-MACAULAY MODULE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Good elements for a maximal regular sequence. Invoked in the counted proof at line 25246. Supplies the case in which the chosen element is a nonzerodivisor with Cohen-Macaulay quotient.
- Cutting down the dimension by choosing a second element. Used in the induction step of the counted proof (lines 25248-25263). Produces an element controlling the quotient so that the induction hypothesis applies.

Source reference. Original lemma 00N4: sources/stacks-algebra/pinned-original-source.tex, source lines 25202–25202. Navigation pointer for \ref{lemma-good-element} used inside the counted span (line 25246); the argument that consumes it is retained in the span.

Source reference. Original lemma 00LJ: sources/stacks-algebra/pinned-original-source.tex, source lines 16856–16856. Navigation pointer for \ref{lemma-permute-xi} used inside the counted span (line 25262); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00DS: sources/stacks-algebra/pinned-original-source.tex, source lines 2781–2781. Navigation pointer for \ref{lemma-silly} used inside the counted span (line 25256); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a Noetherian local ring. Let M be a Cohen-Macaulay module over R. Suppose $g \in \mathfrak{m}$ is such that $\dim(Supp(M) \cap V(g)) = \dim(Supp(M)) - 1$. Then (a) g is a nonzerodivisor on M, and (b) M/gM is Cohen-Macaulay of depth one less.*

Proof. Choose a M-regular sequence $f_1, \ldots, f_d$ with $d = \dim(\text{Supp}(M))$. If g is good with respect to $(M, f_1, \ldots, f_d)$ we win by Lemma 00N4. In particular the lemma holds if $d = 1$. (The case $d = 0$ does not occur.) Assume $d > 1$. Choose an element $h \in R$ such that (i) h is good with respect to $(M, f_1, \ldots, f_d)$, and (ii) $\dim(\text{Supp}(M) \cap V(h, g)) = d - 2$. To see h exists, let $\{\mathfrak{q}_j\}$ be the (finite) set of minimal primes of the closed sets $\text{Supp}(M)$, $\text{Supp}(M) \cap V(f_1, \ldots, f_i)$, $i = 1, \ldots, d-1$, and $\text{Supp}(M) \cap V(g)$. None of these $\mathfrak{q}_j$ is equal to $\mathfrak{m}$ and hence we may find $h \in \mathfrak{m}$, $h \notin \mathfrak{q}_j$ by Lemma 00DS. It is clear that h satisfies (i) and (ii). From Lemma 00N4 we conclude that M/hM is Cohen-Macaulay. By (ii) we see that the pair $(M/hM, g)$ satisfies the induction hypothesis. Hence $M/(h, g)M$ is Cohen-Macaulay and $g : M/hM \to M/hM$ is injective. By Lemma 00LJ we see that $g : M \to M$ and $h : M/gM \to M/gM$ are injective. Combined with the fact that $M/(g, h)M$ is Cohen-Macaulay this finishes the proof. □